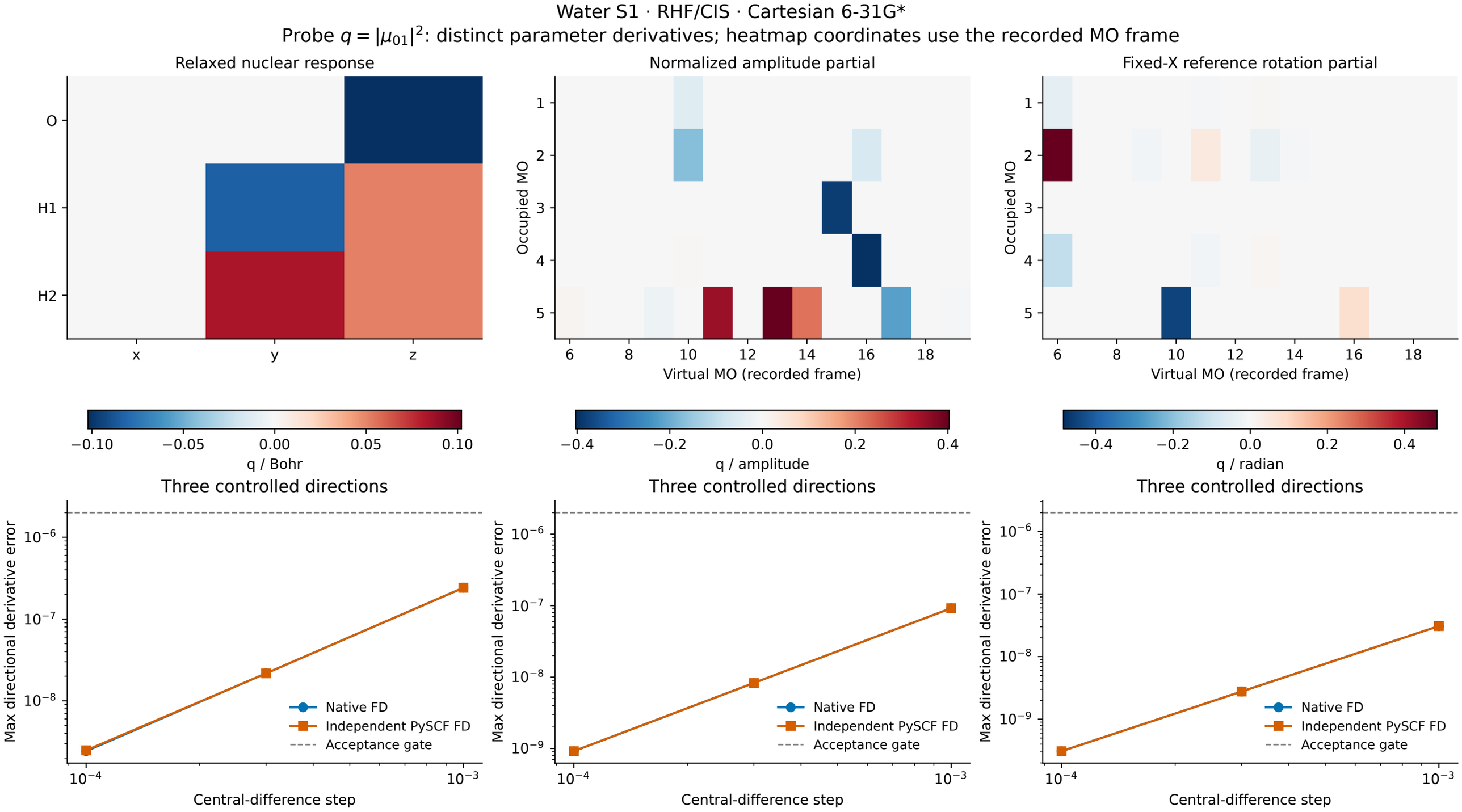
Water S1 · RHF/CIS · Cartesian 6-31G*
Probe $q = |\mu_{01}|^2$: distinct parameter derivatives; heatmap coordinates use the recorded MO frame
Relaxed nuclear response
O
H1
H2
x
y
z
Normalized amplitude partial
Occupied MO
Virtual MO (recorded frame)
Fixed-X reference rotation partial
Occupied MO
Virtual MO (recorded frame)
−0.10
−0.05
0.00
0.05
0.10
q / Bohr
−0.4
−0.2
0.0
0.2
0.4
q / amplitude
−0.4
−0.2
0.0
0.2
0.4
q / radian
Three controlled directions
Max directional derivative error
Native FD
Independent PySCF FD
Acceptance gate
Central-difference step
Three controlled directions
Max directional derivative error
Native FD
Independent PySCF FD
Acceptance gate
Central-difference step
Three controlled directions
Max directional derivative error
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Central-difference step